# Midterm Project

**Simple Player Management System**

**Explanation**: This program simulates an admin panel that is used to manage players inside of a video game. Players have certain information and attributes including string name, int level, int id, array of 5 stats, int skill points.

The admin panel allows the user to:

1. Add a player given name, level, id
2. Remove a player given id
3. Display all players
4. Search a player by id
5. Export all players' data in the system into a csv file
6. Import players' data from a csv file
7. Manage a player's attribute by allocating skill points given stat and number of points
8. Exit program

**List of Concepts**

1. Data types and data sizes for storing player information such as name and level
2. File types used to import and export player data using .csv
3. Pointers used to get memory address of player objects stored in a vector in order to maintain program efficiency and to update player statistics
4. Binary search used to search for a specific player in a vector sorted by id
5. Strings used for storing player name and getting user input
6. File I/O with fstream to import and export player data
7. Two classes including Player class and an Admin class, inherited from Player

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**Screenshots**

1. Adding 2 players to the program.

```
Admin Menu:
1. Add Player
2. Remove Player
3. Display All Players
4. Search Player by ID
5. Export to CSV
6. Import from CSV
7. Manage Player Attributes
8. Exit
Enter your choice: 1

Enter player name: Playe1
Name contains invalid characters. Please enter a valid name.
Player1
Name contains invalid characters. Please enter a valid name.
Player One
Enter player level: 10
Enter player ID: 1
Successfully exported player data to players.csv
Player added successfully! (Auto saved to players.csv)


Admin Menu:
1. Add Player
2. Remove Player
3. Display All Players
4. Search Player by ID
5. Export to CSV
6. Import from CSV
7. Manage Player Attributes
8. Exit
Enter your choice: 1

Enter player name: Player Two
Enter player level: 10
Enter player ID: 2
Successfully exported player data to players.csv
Player added successfully! (Auto saved to players.csv)
```

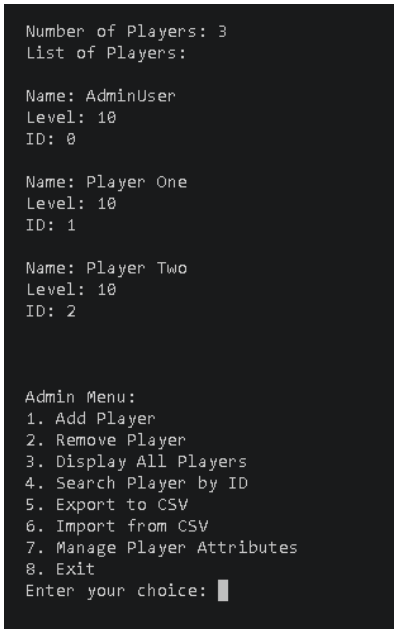
2. Automatically creates and updates to players.csv

Midterm > players.csv

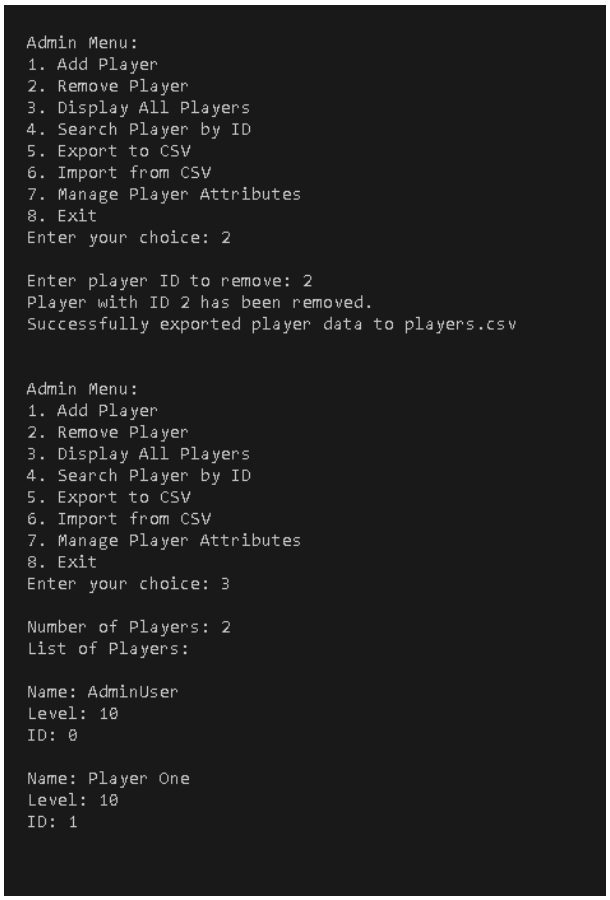
```
1   ID,Name,Level,Vitality,Strength,Agility,Intellegience,Sense,SkillPoints
2   1,Player One,10,10,10,10,10,10,30
3   2,Player Two,10,10,10,10,10,10,30
4
```

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3. Showing all players

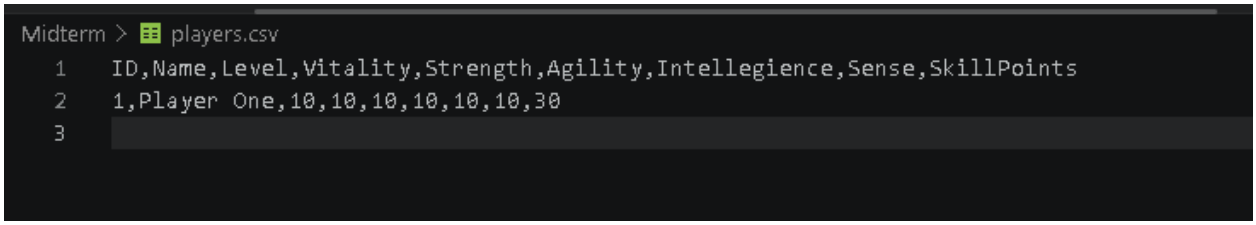



4. Removing player two

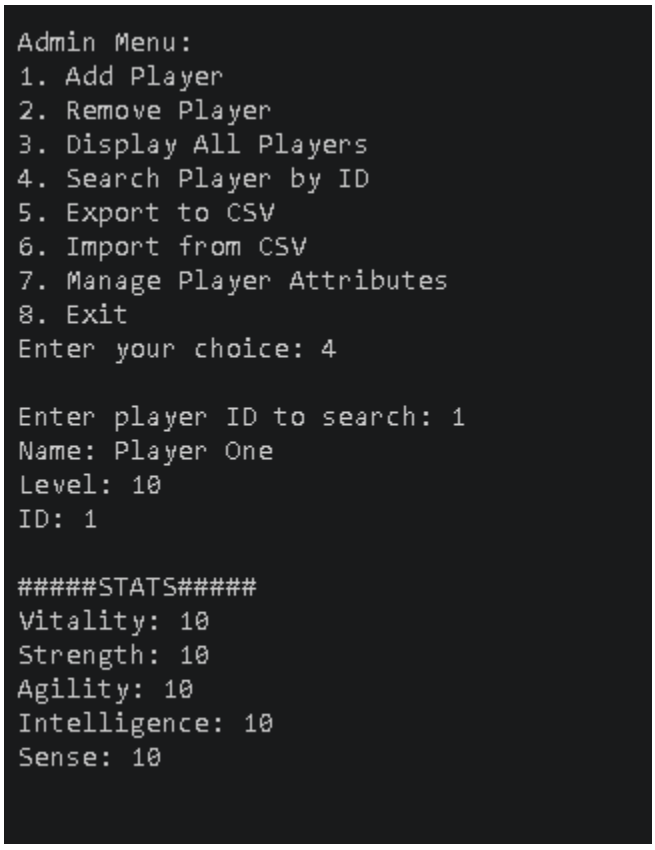


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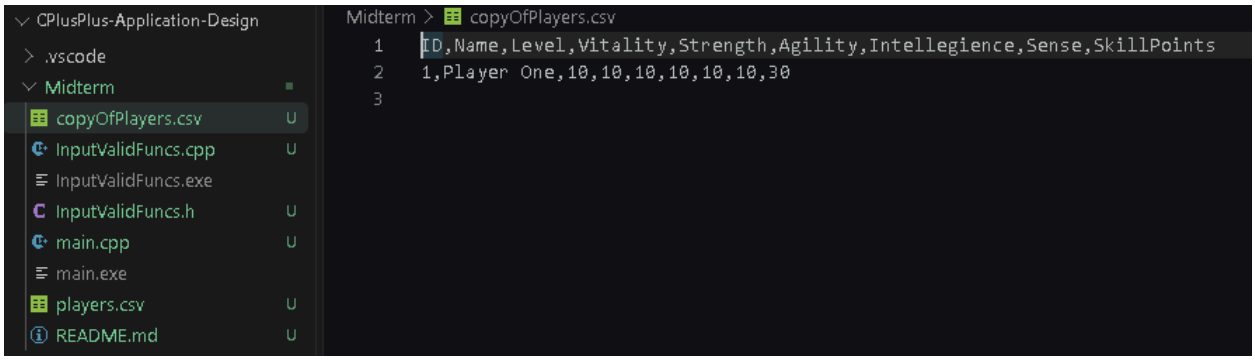
5. players.csv updates automatically for the removal of player two




6. Search and shows player statistics given player id




7. Export to another data onto another .csv file



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8. Restarts program and import the new csv file into the system

```
p -o main } ; if ($?) { .\main }

Admin Menu:
1. Add Player
2. Remove Player
3. Display All Players
4. Search Player by ID
5. Export to CSV
6. Import from CSV
7. Manage Player Attributes
8. Exit
Enter your choice: 6

Enter CSV filename to import from (e.g., players): copyOfPlayers
Successfully imported 1 player(s) from copyOfPlayers.csv


Admin Menu:
1. Add Player
2. Remove Player
3. Display All Players
4. Search Player by ID
5. Export to CSV
6. Import from CSV
7. Manage Player Attributes
8. Exit
Enter your choice: 3

Number of Players: 2
List of Players:

Name: AdminUser
Level: 10
ID: 0

Name: Player One
Level: 10
ID: 1
```